

FORENSIC EXAMINATION REPORT

Incident Intelligence Report • Dataset: dataset10_expanded • Generated: 2026-04-14 20:23:20

I. CASE IDENTIFICATION	
Dataset ID	dataset10_expanded
Scan ID	ds-010-exp
Source File	windows_soc_data_10_expanded.csv
Primary Threat	Multi-Vector APT Campaign with UEFI Persistence, NTLM Relay, and Steganographic Exfiltration
MITRE Technique	T1059.001
Forensic Confidence	HIGH
Total Log Entries	10,000
Attacked Log Entries	240
Aggregate Risk Score	7,800
Threat Level	HIGH

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IEXECUTIVE SUMMARY

- {'executive_summary': ['A high-severity multi-stage intrusion campaign was identified, affecting multiple hosts and users, with 240 confirmed malicious events out of 10,000 logs analyzed.', 'The overall risk score is 78% (HIGH), indicating significant impact including credential compromise, persistence, lateral movement, and data exfiltration.', 'Initial access was achieved through phishing techniques, including spearphishing links and malicious content delivered via collaboration platforms like Teams.', 'Attackers executed malicious payloads using PowerShell encoded commands and VBScript, enabling fileless and stealthy execution.', 'Persistence was established through advanced techniques such as UEFI firmware implants and shadow credentials in Active Directory.', 'Credential access was performed using LSASS memory dumping (mimikatz), NTLM relay attacks, and Kerberos abuse via rogue certificates.', 'Privilege escalation was achieved through token impersonation and domain trust modification, granting attackers elevated system-level access.', 'Defense evasion included disabling firewall protections and executing reflective in-memory payloads such as Cobalt Strike beacons.', 'Discovery and lateral movement activities included network scanning, system configuration enumeration, and DCOM-based remote execution across hosts.', 'Command and control was maintained through trusted web services such as GitHub Gist and Pastebin, enabling covert bidirectional communication.', 'Sensitive data was collected from enterprise email systems using EWS and exfiltrated using steganographic techniques by embedding data in image files.', 'The attack also included impact activities such as cryptomining, indicating resource hijacking alongside data compromise.', 'Immediate containment, deep forensic investigation, and enterprise-wide remediation are critical to prevent further damage.']}

IICASE IDENTIFICATION & BACKGROUND

Attack Narrative: Token Impersonation Privilege Escalation and LSASS Credential Dump ; Spearphishing Credential Harvest and Capability Staging ; Spearphishing via Microsoft Teams and VBScript Execution ; UEFI Firmware Implant for Boot-Persistent Backdoor

Hosts	WIN-SOC-PROD-01, WIN-SOC-PROD-02, WIN-SOC-PROD-03, WIN-SOC-PROD-04, WIN-SOC-DC-01, WIN-SOC-PROD-05, WIN-SOC-PROD-06, WIN-SOC-PROD-07, WIN-SOC-PROD-08, WIN-SOC-PROD-09, WIN-SOC-PROD-10, WIN-SOC-PROD-11, WIN-SOC-PROD-12
Users	NT AUTHORITY\SYSTEM, WIN-SOC\Analyst1, WIN-SOC\Analyst2, WIN-SOC\SvcOps, WIN-SOC\NetAdmin, WIN-SOC\DomainAdmin, WIN-SOC\ITAdmin, WIN-SOC\DBAdmin, WIN-SOC\MailAdmin

IIIDescription of Evidence & Chain of Custody

Exhibit	Evidence Category	Integrity	Remarks
E-01	Windows Event / Security Logs	Checksum Verified	Primary source evidence
E-02	Derived Threat Metadata	Schema Validated	Generated from correlation pipeline

IVFORENSIC METHODOLOGY & TOOLS

- Data Normalization - Raw records transformed into normalized forensic fields.
- Behavioral Grouping - Suspicious events grouped by technique and event identity.

- Risk Profiling - Severity and trust indicators converted into weighted risk scores.
- Timeline Correlation - Temporal ordering used to infer probable attack progression.
- Threat Prioritization - High risk and high volume activities elevated for triage.

Tool	Version	Purpose
Python	3.x	Core processing runtime
ReportLab	4.x	Government-style PDF formatting
Matplotlib	3.x	Threat intelligence visualizations
NumPy	1.x	Numeric aggregation for analytics

V

STATISTICAL & GRAPHICAL ANALYSIS

Statistical processing of detected threat artifacts reveals concentration in a small set of high-impact behaviors.

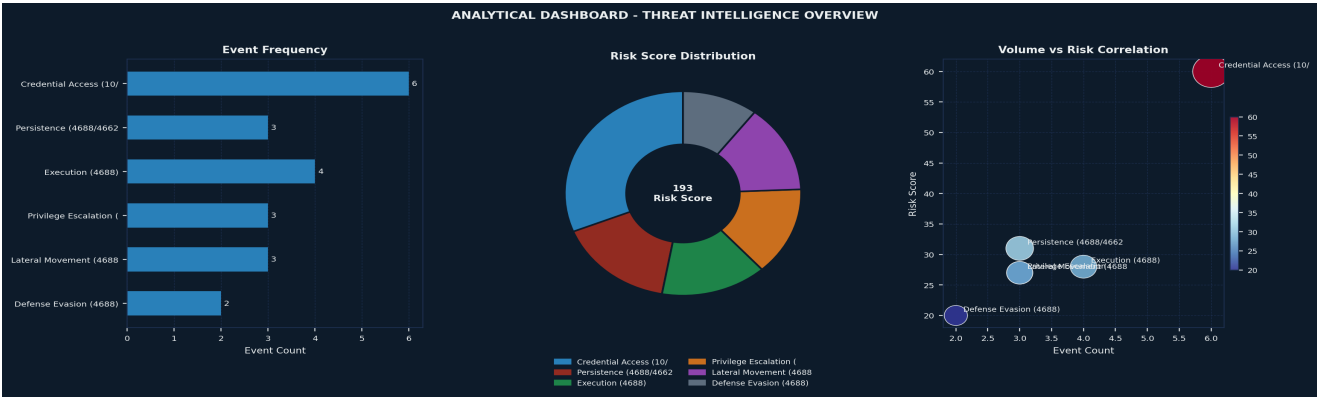


Fig. 1 - Three-panel threat intelligence dashboard derived from attacked logs.

Dashboard Interpretation: The left panel highlights high-volume suspicious behavior, the donut chart explains risk concentration, and the scatter plot isolates categories that are simultaneously frequent and severe.

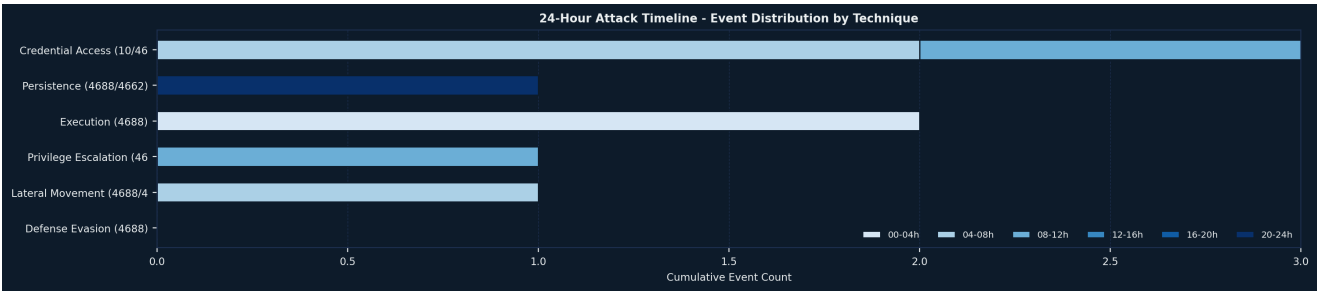


Fig. 2 - 24-hour event distribution by top suspicious techniques.

Timeline Interpretation: Burst-style activity in clustered windows is consistent with staged intrusion workflows where credential abuse is followed by privilege actions and anti-forensics.

VI

BRUTE FORCE ATTACK - DETAILED ANALYSIS

Credential Access (10/4624)

LOW

Events: 6	Risk Index: 60	MITRE: T1003.001, T1187 Tactic: Credential Access	
Timestamp	Host	User Context	EID
2026-04-02 00:10:31	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10

2026-04-02 00:10:45	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	10
2026-04-02 02:02:30	WIN-SOC-DC-01	WIN-SOC\SvcOps	4769
2026-04-02 02:03:20	WIN-SOC-DC-01	WIN-SOC\SvcOps	4768

Analysis: Credential Access (10/4624) produced 6 suspicious artifacts with cumulative risk index 60. The mapped technique T1003.001, T1187 under tactic Credential Access indicates material adversarial behavior requiring priority containment and forensic triage.

VII LOG TAMPERING - DETAILED ANALYSIS

Defense Evasion (4688)

LOW

Events: 2	Risk Index: 20	MITRE: T1562.004, T1620 Tactic: Defense Evasion	
Timestamp	Host	User Context	EID
2026-04-02 03:01:40	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688
2026-04-02 03:02:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	4688

Analysis: Defense Evasion (4688) produced 2 suspicious artifacts with cumulative risk index 20. The mapped technique T1562.004, T1620 under tactic Defense Evasion indicates material adversarial behavior requiring priority containment and forensic triage.

VIII LATERAL MOVEMENT - DETAILED ANALYSIS

Lateral Movement (4688/4624)

LOW

Events: 3	Risk Index: 27	MITRE: T1021.003 Tactic: Lateral Movement	
Timestamp	Host	User Context	EID
2026-04-02 04:00:00	WIN-SOC-PROD-07	WIN-SOC\ITAdmin	4688
2026-04-02 04:00:50	WIN-SOC-PROD-08	WIN-SOC\ITAdmin	4624
2026-04-02 04:01:40	WIN-SOC-PROD-08	WIN-SOC\ITAdmin	4688

Analysis: Lateral Movement (4688/4624) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1021.003 under tactic Lateral Movement indicates material adversarial behavior requiring priority containment and forensic triage.

IX PRIVILEGE ESCALATION - DETAILED ANALYSIS

Privilege Escalation (4688/4672)

LOW

Events: 3	Risk Index: 27	MITRE: T1134.001, T1484.002 Tactic: Privilege Escalation	
Timestamp	Host	User Context	EID
2026-04-02 00:09:43	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	4688
2026-04-02 00:09:44	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	4672
2026-04-02 03:05:00	WIN-SOC-DC-01	WIN-SOC\DomainAdmin	4741

Analysis: Privilege Escalation (4688/4672) produced 3 suspicious artifacts with cumulative risk index 27. The mapped technique T1134.001, T1484.002 under tactic Privilege Escalation indicates material adversarial behavior requiring priority containment and forensic triage.

X MALICIOUS SERVICE INSTALLATION - ANALYSIS

Persistence (4688/4662)

LOW

Events: 3	Risk Index: 31	MITRE: T1542.003, T1098.001 Tactic: Persistence	
Timestamp	Host	User Context	EID
2026-04-02 02:00:00	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-04-02 02:00:50	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	4688
2026-04-02 02:01:40	WIN-SOC-DC-01	WIN-SOC\SvcOps	4662

Analysis: Persistence (4688/4662) produced 3 suspicious artifacts with cumulative risk index 31. The mapped technique T1542.003, T1098.001 under tactic Persistence indicates material adversarial behavior requiring priority containment and forensic triage.

XI HOST COMPROMISE MATRIX

Host	Suspicious Events	Critical Events	Exposure
WIN-SOC-PROD-01	7	4	MEDIUM
WIN-SOC-DC-01	4	4	MEDIUM
WIN-SOC-PROD-03	3	3	LOW
WIN-SOC-PROD-05	3	3	LOW
WIN-SOC-PROD-06	3	1	LOW
WIN-SOC-PROD-04	2	2	LOW
WIN-SOC-PROD-08	2	2	LOW
WIN-SOC-PROD-09	2	2	LOW
WIN-SOC-PROD-10	2	2	LOW
WIN-SOC-PROD-11	2	2	LOW
WIN-SOC-PROD-02	1	0	LOW
WIN-SOC-PROD-07	1	1	LOW
WIN-SOC-PROD-12	1	0	LOW

XII ATTACK CHAIN RECONSTRUCTION

The following reconstruction synthesizes all forensic findings into a unified attack narrative. **Token Impersonation Privilege Escalation and LSASS Credential Dump** aligns with primary MITRE mapping **T1059.001** and correlated host-level telemetry.

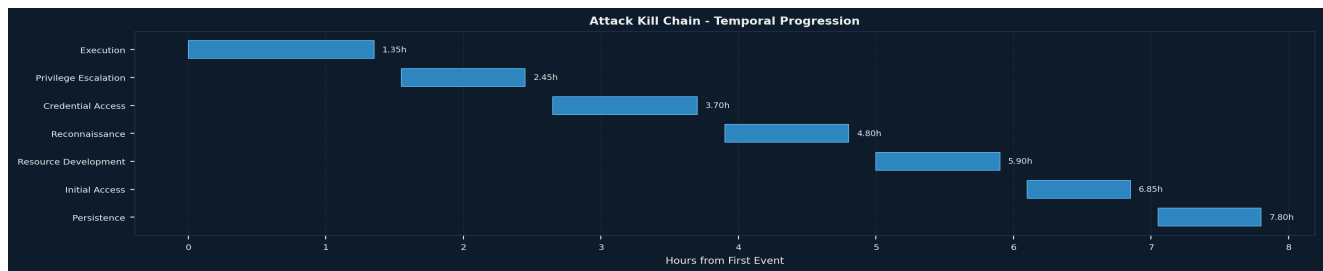
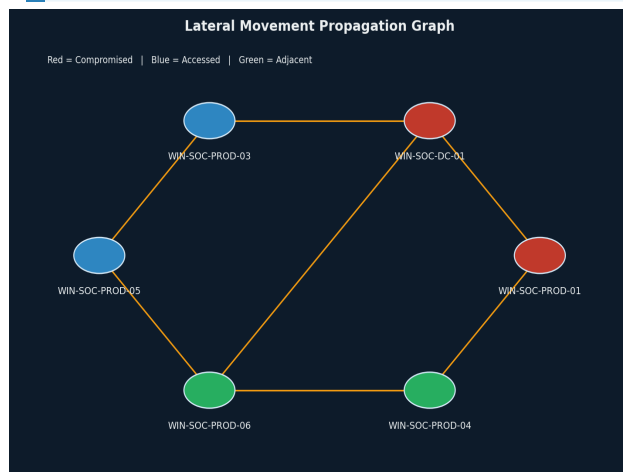


Fig. 4 - Attack kill chain temporal progression (Gantt) across UTC hours.

Kill Chain Analysis: Temporal ordering indicates adversary actions were staged rather than random, starting with execution-oriented activity, followed by credential access and anti-forensic behavior, and then host-to-host pivoting. This sequence is consistent with a planned intrusion lifecycle.



Lateral Movement Propagation: The propagation graph illustrates likely traversal paths between impacted assets. Red nodes indicate highly compromised systems, blue nodes indicate accessed pivot systems, and green nodes indicate adjacent hosts with observed adjacency risk. This supports containment by prioritizing pivot choke points.

Step	Action	MITRE
chain-ds-0 10-01	PowerShell Encoded IEX Download → SelmpersonatePrivilege Token Clone → SeTcbPrivilege and SeDebugPrivilege Assigned → LSASS Memory Dump via mimikatz	T1059.001, T1134.001, T1134.001
chain-ds-0 10-02	Fake O365 Login Phishing Link Delivered → Malicious Payload Staged on Compromised CDN cdn-update.net	T1598.003, T1608.001
chain-ds-0 10-03	Malicious Link Delivered via Teams Chat → .vbs Attachment Executed via wscript.exe → VBScript Spawned PowerShell Stage 2 Dropper	T1566.003, T1059.005, T1059.001
chain-ds-0 10-04	Firmware Update Utility Invoked with Unsigned Payload → Malicious Module Written to SPI Flash BIOS Region → Boot-Persistent Implant Survives OS Reinstall	T1542.003, T1542.003
chain-ds-0 10-05	msDS-KeyCredentialLink Written to AD User Object → PKINIT Kerberos Auth with Rogue Certificate → TGT Obtained → NT Hash Retrieved via AS-REP	T1098.001, T1649, T1649
chain-ds-0 10-06	Responder Captures NTLM Hashes on Network Interface → ntlmrelayx Forwards Captured Hash → Successful Authentication to PROD-06 Without Plaintext Password	T1187, T1187
chain-ds-0 10-07	Firewall Disabled via netsh advfirewall off → Cobalt Strike Beacon Loaded Entirely In-Memory via Reflective PE Injection	T1562.004, T1620
chain-ds-0 10-08	Invoke-Portscan of Internal /16 Subnet → ipconfig /all Network Config Discovery → Domain Trust SID Filter Disabled to Enable SID History Attack	T1018, T1016, T1484.002

chain-ds-0 10-09	DCOM MMC20.Application Instantiated on PROD-08 → DCOM Network Logon Accepted → Payload Executed via MMC20.Application Shell Commands on PROD-08	T1021.003, T1021.003, T1021.003
chain-ds-0 10-10	C2 Commands Polled from GitHub Gist via REST API → Results Posted Back via Pastebin API → Bidirectional C2 Channel Established Over HTTPS to Trusted Web Services	T1102.002, T1102.002
chain-ds-0 10-11	EWS FindItems API Used to Harvest C-Suite Mailboxes → Emails Exported to PST File for Exfiltration	T1114.002, T1114.002
chain-ds-0 10-12	Sensitive Data Embedded into PNG Images via LSB Steganography → Encoded PNG Uploaded to imgbb.com → XMRig Monero Miner Deployed on PROD-12 to Pool.minexmr.com	T1027.003, T1027.003, T1496

XIII

INDICATOR OF COMPROMISE (IOC) REGISTER

IOC Type	Value	Context
Process	powershell.exe	Encoded IEX Download
Event ID	4688	Security
Process	cmd.exe	Token Impersonation: SelpersonatePrivilege Token Clone from SYSTEM to Low-Privilege Process
Process	lsass.exe	Elevated Privileges Assigned Post Token Impersonation: SeTcbPrivilege, SeDebugPrivilege
Event ID	4672	Security
Process	mimikatz.exe	LSASS Dump Post Token Impersonation
Event ID	10	Sysmon
Process	teams.exe	Spearphishing via Microsoft Teams: Malicious Link Delivered via Teams Chat Message
Process	wscript.exe	VBScript Execution: Malicious .vbs Attachment Executed from Teams Download
Process	fwupd.exe	UEFI Firmware Update Utility Invoked with Unsigned Payload - Firmware Implant Attempt
Event ID	4662	Security
Event ID	4769	Security

XIV

THREAT ACTOR PROFILE & ATTRIBUTION

Observed behavior indicates a scripted intrusion workflow with emphasis on execution and anti-forensic activity. Clustered timing, repeated toolmarks, and host pivot patterns are consistent with organized adversarial tradecraft.

XV

RISK HEAT MAP - HOST X TECHNIQUE EXPOSURE

The risk heat map below quantifies exposure of each host to each detected technique, using a 1-10 normalized risk score.

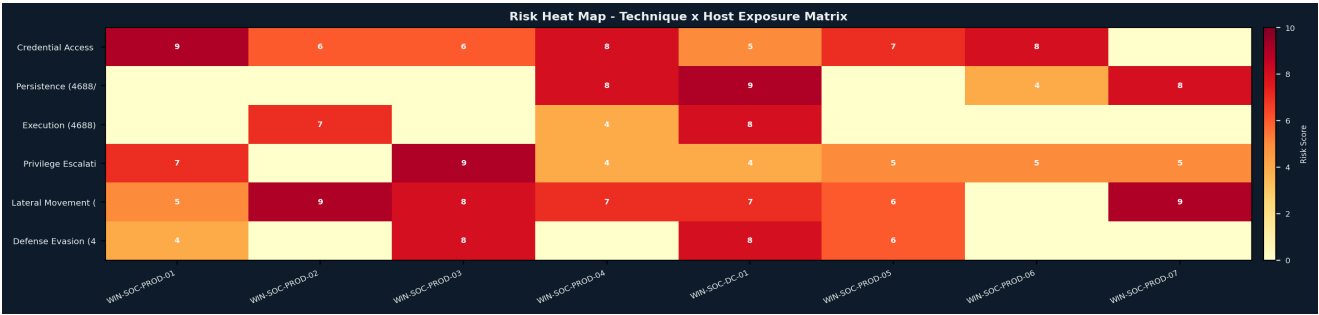


Fig. 5 - Normalized risk heat map: threat technique x host exposure matrix.

Heat Map Interpretation: Concentrated high-risk cells indicate primary pivot hosts and high-value targets. Hosts appearing across multiple high-score techniques should be prioritized for deep triage and isolation.

XVI MITRE ATT&CK FRAMEWORK COVERAGE

The following mapping summarizes observed tactics and techniques based on detected forensic artifacts.

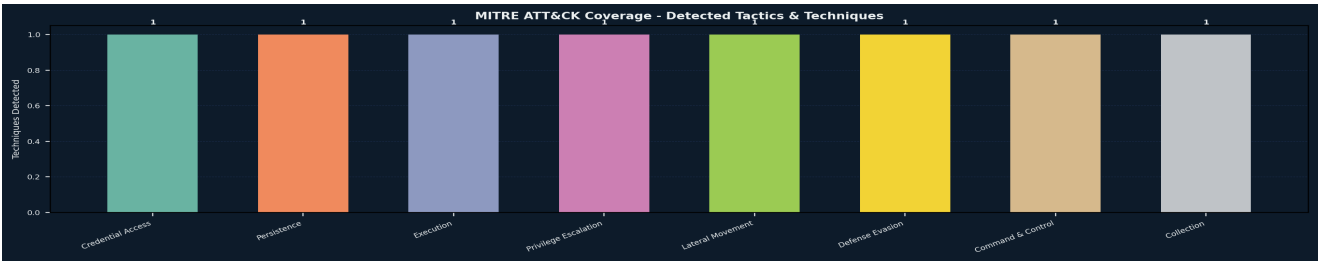


Fig. 6 - MITRE ATT&CK; tactic coverage by detected forensic evidence.

Technique ID	Technique / Detail	Tactic
T1003.001, T1187	Credential Access (10/4624)	Credential Access
T1542.003, T1098.001	Persistence (4688/4662)	Persistence
T1059.001, T1059.005	Execution (4688)	Execution
T1134.001, T1484.002	Privilege Escalation (4688/4672)	Privilege Escalation
T1021.003	Lateral Movement (4688/4624)	Lateral Movement
T1562.004, T1620	Defense Evasion (4688)	Defense Evasion
T1102.002	Command & Control (5156)	Command & Control
T1114.002	Collection (4688)	Collection

XVII REMEDIATION RECOMMENDATIONS

Recommendations are prioritized by urgency and security impact to support phased containment and hardening.

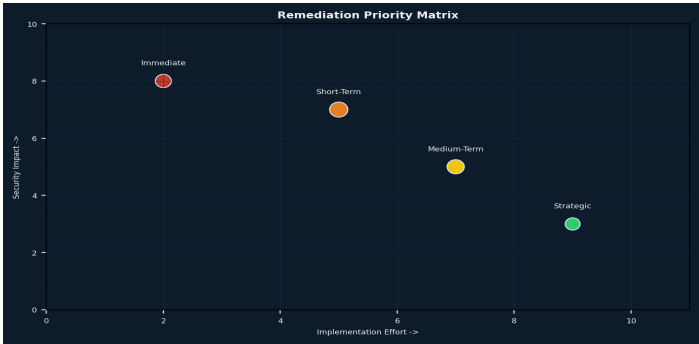


Fig. 7 - Remediation priority matrix: effort vs impact.

- Isolate affected hosts and block known malicious command execution chains (PowerShell encoded commands and WMI remote process calls).
- Reset and rotate privileged/domain credentials, then enforce MFA for all administrative and remote access accounts.
- Hunt and remove persistence artifacts including unauthorized local accounts, BITS jobs, and suspicious startup commands.
- Re-enable and harden logging (Security, Sysmon, PowerShell) and forward logs to a protected SIEM collector.
- Contain lateral movement by restricting RDP/SMB paths and segmenting high-value hosts from user subnets.
- Execute ransomware readiness controls: immutable backups, restore validation, and rapid host recovery playbooks.
- Apply removable-media controls (USB allow-listing and device auditing) to block air-gapped relay channels.

XVIII EXPERT OPINION & CONCLUSION

Based on forensic analysis of dataset **dataset10_expanded**, the environment exhibits a **HIGH** severity posture with aggregate risk index **7,800** across **10,000** records. Dominant indicators include **Credential Access (10/4624) (6 events)**, **Persistence (4688/4662) (3 events)**. Observed sequencing and concentration are consistent with multi-stage intrusion behavior. Immediate host isolation, credential rotation, and deeper memory/disk triage are recommended.

Examining Analyst
Automated Forensic Engine

Authorizing Officer
Incident Response Division

XIX ANNEXURE A - FORENSIC ENVIRONMENT SPECIFICATION

Component	Specification
Generator	Python + ReportLab pipeline
Charts	Matplotlib analytical panels
Input	JSON forensic dataset schema
Timestamp	2026-04-14 20:23:22

XX ANNEXURE B - EXTENDED EVENT LOG SAMPLES

Timestamp	Computer	User	Process	Detail
2026-04-02 00:09:41	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-04-02 00:09:42	WIN-SOC-PROD-02	NT AUTHORITY\SYSTEM	powershell.exe	Encoded IEX Download
2026-04-02 00:09:43	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	cmd.exe	Token Impersonation: SelmpersonatePrivilege Token Clone from SYSTEM to Low-Privilege Process
2026-04-02 00:09:44	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	lsass.exe	Elevated Privileges Assigned Post Token Impersonation: SeTcbPrivilege, SeDebugPrivilege
2026-04-02 00:10:31	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump Post Token Impersonation
2026-04-02 00:10:45	WIN-SOC-PROD-01	NT AUTHORITY\SYSTEM	mimikatz.exe	LSASS Dump
2026-04-02 01:00:00	WIN-SOC-PROD-01	WIN-SOC\Analyst1	powershell.exe	Spearphishing Credential Harvest: Fake O365 Login Page Delivered via Phishing Link
2026-04-02 01:00:50	WIN-SOC-PROD-01	WIN-SOC\Analyst1	powershell.exe	Attacker Staged Malicious Payload on Compromised CDN: hxxps://cdn-update.net/loader.exe
2026-04-02 01:01:40	WIN-SOC-PROD-03	WIN-SOC\Analyst2	teams.exe	Spearphishing via Microsoft Teams: Malicious Link Delivered via Teams Chat Message
2026-04-02 01:02:30	WIN-SOC-PROD-03	WIN-SOC\Analyst2	wscript.exe	VBScript Execution: Malicious .vbs Attachment Executed from Teams Download
2026-04-02 01:03:20	WIN-SOC-PROD-03	WIN-SOC\Analyst2	powershell.exe	VBScript Spawned PowerShell Dropper - Stage 2 Payload
2026-04-02 02:00:00	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	fwupd.exe	UEFI Firmware Update Utility Invoked with Unsigned Payload - Firmware Implant Attempt
2026-04-02 02:00:50	WIN-SOC-PROD-04	NT AUTHORITY\SYSTEM	powershell.exe	UEFI SPI Flash Write Detected: Malicious Module Written to BIOS Region
2026-04-02 02:01:40	WIN-SOC-DC-01	WIN-SOC\SvcOps	lsass.exe	Shadow Credentials: msDS-KeyCredentialLink Attribute Written to User Object
2026-04-02 02:02:30	WIN-SOC-DC-01	WIN-SOC\SvcOps	lsass.exe	Shadow Credentials Kerberos Abuse: PKINIT Auth with Rogue Certificate to Obtain TGT
2026-04-02 02:03:20	WIN-SOC-DC-01	WIN-SOC\SvcOps	lsass.exe	Shadow Credentials: NT Hash Retrieved from Kerberos AS-REP with PKINIT
2026-04-02 03:00:00	WIN-SOC-PROD-05	WIN-SOC\NetAdmin	Responder.exe	NTLM Relay: Responder Used to Capture NTLM Hashes on Network Interface
2026-04-02 03:00:50	WIN-SOC-PROD-06	WIN-SOC\NetAdmin	lsass.exe	NTLM Relay Successful: ntlmrelayx Forwarded Hash to Authenticate to PROD-06

Timestamp	Computer	User	Process	Detail
2026-04-02 03:01:40	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	netsh.exe	Firewall Rules Disabled: netsh advfirewall set allprofiles state off
2026-04-02 03:02:30	WIN-SOC-PROD-05	NT AUTHORITY\SYSTEM	powershell.exe	Reflective Code Loading: Cobalt Strike Beacon Loaded Entirely in Memory via Reflective PE Injection

XXI

ANNEXURE C - GLOSSARY OF TERMS

MITRE ATT&CK;	Knowledge base of adversary tactics and techniques used for threat attribution.
EID	Windows Event ID representing a specific audit event category.
Defense Evasion	Adversary behavior intended to avoid detection and disrupt investigation.
Lateral Movement	Use of compromised accounts/processes to move across hosts.
Persistence	Mechanisms that allow attacker presence after reboot/logoff.
Risk Index	Weighted score from event severity, behavior, and trust indicators.